SURVEILLANCE FOR CRIME PREVENTION IN PUBLIC SPACE: RESULTS AND POLICY CHOICES IN BRITAIN AND AMERICA

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Research Summary:

The present study assessed the comparative effectiveness of two forms of surveillance in preventing crime in public space: formal surveillance, in the form of closed-circuit television (CCTV) surveillance cameras, and natural surveillance, in the form of improved street lighting. Based on the highest quality available research evidence on these two measures—from systematic reviews incorporating meta-analytic techniques—it was found that they are equally effective in reducing crime. More detailed analyses showed that improved street lighting was more effective in reducing crime in city centers, that both were more effective in reducing property crimes than violent crimes, and that both measures were far more effective in reducing crime in Britain than in America.

Policy Implications:

The results of this study lend support for the continued use of CCTV and improved street lighting to reduce crime, but they suggest that they need to be targeted more narrowly on property crimes. A policy that combines the two interventions may produce a greater yield in reduced crime rates, but this may be limited to vehicle crimes in car parks. Improved street lighting is an effective form of surveillance to reduce crime in public space, and it may attract less public resistance than CCTV surveillance cameras (in America at least). Lessons from the British studies to help improve the effectiveness of the two measures in America are explored.

KEYWORDS: Crime Prevention, Surveillance, Closed-Circuit Television (CCTV), Improved Street Lighting, Systematic Review, Meta-Analysis

For many Americans, the idea of some form of surveillance technology, such as CCTV cameras, being used in public areas conjures up images of Orwell's "Big Brother" society—a society that is constantly watching (and recording) every move, every action that one takes. According to a recent report of the American Civil Liberties Union, America may be on the

verge of becoming a "surveillance society" (Stanley and Steinhardt, 2003:1, emphasis in original):

> The explosion of computers, cameras, sensors, wireless communication, GPS, biometrics, and other technologies in just the last 10 years is feeding a surveillance monster that is growing silently in our midst. Scarcely a month goes by in which we don't read about some new high-tech way to invade people's privacy, from face recognition to implantable microchips, data-mining, DNA chips, and even "brain wave fingerprinting." The fact is, there are no longer any *technical* barriers to the Big Brother regime portrayed by George Orwell.

But not all forms of surveillance are as potentially intrusive and raise questions about the infringement of civil liberties. Indeed, surveillance for the purpose of preventing crime can mean many different things. Clarke and Homel (1997), in their revised classification of situational crime prevention, differentiate among three types of surveillance, each aimed at increasing offenders' perceived risks of committing a crime: formal surveillance, natural surveillance, and surveillance by employees (i.e., by virtue of their position, they perform a surveillance function, e.g., parking lot attendants, bus drivers). The main aim of this paper is to assess the comparative effectiveness of two of these three forms of surveillance in preventing "traditional" crimes (e.g., robbery, burglary, motor vehicle crimes) in public space: formal surveillance, in the form of CCTV surveillance cameras, and natural surveillance, in the form of improved street lighting.

Formal surveillance aims to produce a "deterrent threat to potential offenders" through the deployment of personnel whose primary responsibility is security (e.g., police, security guards) or through the introduction of some form of technology, such as CCTV, to enhance or take the place of security personnel (Clarke, 1997:20). Natural surveillance shares the same aim as formal surveillance, but it involves efforts that seek to "capitalize upon the 'natural' surveillance provided by people going about their everyday business." Examples of natural surveillance include the installation or improvement of street lighting, the removal of objects from windows of convenience stores that obscure lines of sight into the store, and the removal or pruning of bushes in front of homes so residents may have a clear view of outside (Clarke, 1997:21).

Why compare the effectiveness of CCTV surveillance cameras and improved street lighting in preventing crime? There are two main interests, one of a historical and the other of a contemporary nature. The more historical interest pertains to the widespread use of improved street lighting in the United States in the 1970s and its subsequent abandonment as a policy option. After the dramatic increase in crime in the 1960s, many

American towns and cities embarked on major street lighting programs as a means of reducing crime. This led to a detailed review of the effectiveness of street lighting in preventing crime by Tien et al. (1979), as part of the National Evaluation Program of Law Enforcement Assistance Administration funding. On the basis of the 15 most methodologically rigorous evaluations, Tien et al. (1979) concluded that the results of street lighting on crime were mixed and generally inconclusive.

Just as Robert Martinson's (1974) review of offender rehabilitation programs was interpreted as showing that nothing works, Tien et al.'s (1979) review was interpreted as showing that street lighting had no effect on crime. However, both reviews relied on what are now called "vote-counting" methods, focusing on how many evaluations produced statistically significant results. Unfortunately, statistical significance depends partly on the effect size and partly on the sample size. Many criminological evaluations are based on relatively small numbers, which means that a particularly important effect size may not be statistically significant. Vote-counting methods were supplanted in the 1980s by meta-analysis, which focuses on the effect size and estimates of weighted mean effect sizes of interventions (Lipsey and Wilson, 2001). If Martinson (1974) and Tien et al. (1979) had carried out meta-analyses, it seems likely that they would have reached different conclusions.

Tien et al.'s (1979) review effectively ended research on street lighting and crime in the United States. And although offender rehabilitation did not endure the same long-lasting fate as street lighting, these two crime prevention policies, on the basis of the findings of the respective reviews, shared the same distinction of being judged erroneously. In the case of street lighting, because the review found the evidence to be mixed and generally inconclusive (many of the programs were effective), this should have led instead to attempts to evaluate the effects of street lighting using more adequate designs and should have stimulated efforts to determine in what circumstances improved street lighting might lead to reductions in crime.

The contemporary interest in comparing these two situational crime prevention measures arises largely from the current British experience of the widespread use of one of these measures (CCTV) as a government-funded method of reducing crime and the complete dismissal by the government of the other (improved street lighting). In Britain, CCTV has been and continues to be the most heavily funded noncriminal justice crime prevention measure. Over the three-year period of 1999 to 2001, the British government made available £170 million (approximately $320 million) for "CCTV schemes in town and city centres, car parks, crime hot-spots and residential areas" (Home Office Policing and Reducing Crime Unit, 2001:8). In previous years (1996 to 1998), CCTV accounted

for more than three-quarters of total spending on crime prevention by the British Home Office (Koch, 1998:49). According to Armitage (2002), the number of surveillance cameras increased from 100 in 1990 to 400 in 1994, 5,200 in 1997, to 40,000 in 2002.

There is concern that this funding may have been based partly on political considerations (e.g., the popularity of CCTV with the public) and partly on a handful of apparently successful schemes that were usually evaluated using simple one group (no control group) before-after designs. These evaluations were conducted with varying degrees of competence (Armitage et al., 1999:226) and were often lacking in professional independence from government (Ditton and Short, 1999:202).

Moreover, during this time, there was a complete dismissal by the Home Office of a number of high-quality British studies showing that improved street lighting had a desirable effect on crime. In a recent review of these studies, Pease (1999:68) stated that "the capacity of street lighting to influence crime has now been satisfactorily settled." That substantial funding was poured into CCTV schemes on the basis of questionable research, while an effective alternative in the form of improvements to street lighting—supported by high quality research—was widely known, raises serious questions about the use of public resources to prevent crime in Britain.

HOW MIGHT CCTV AND STREET LIGHTING REDUCE CRIME?

Explanations of the way that CCTV surveillance cameras and improvements in street lighting could prevent crime can be found in situational approaches that focus on reducing opportunity and increasing perceived risk through modification of the physical environment (Clarke, 1995), and in perspectives that stress the importance of strengthening informal social control and community cohesion through more effective street use (Jacobs, 1961) and investment in neighborhood conditions (Taub et al., 1984; Taylor and Gottfredson, 1986).

The situational approach to crime prevention suggests that crime can be prevented by environmental measures that directly affect offenders' perceptions of increased risks and decreased rewards. This approach is also supported by theories that emphasize natural, informal surveillance as a key to crime prevention. For example, Jacobs (1961) drew attention to the role of good visibility combined with natural surveillance as a deterrent to crime. She emphasized the association between levels of crime and public street use, suggesting that less crime would be committed in areas with an abundance of potential witnesses.

Other theoretical perspectives have emphasized the importance of

investment to improve neighborhood conditions as a means of strengthening community confidence, cohesion, and social control (Skogan, 1990; Wilson and Kelling, 1982). Sampson et al. (1997) argued that a low degree of "collective efficacy" in a neighborhood (a low degree of informal social control) caused high crime rates. As highly visible signs of investment, CCTV and improved street lighting might reduce crime if they physically improved the environment and signaled to residents that efforts were being made to invest in and improve their neighborhood. In turn, this might lead residents to have a more positive image of the area and increased community pride, optimism, and cohesion. In turn, this might lead residents to exert greater informal social control over potential offenders in an area. It should be noted that this theoretical perspective predicts a reduction in both daytime and nighttime crime. Consequently, attempts to measure the effects of improved street lighting on crime should not concentrate purely on nighttime crime. The same is true for evaluations of CCTV because infrared and other night vision technology allows CCTV to operate just as effectively during nighttime as during daytime.

The relationship among visibility, social surveillance, and criminal opportunities is a consistently strong theme to emerge from the literature. A core assumption of both opportunity and informal social control models of prevention is that criminal opportunities and risks are influenced by environmental conditions in interaction with resident and offender characteristics. Street lighting and CCTV are tangible alterations to the built environment, but they do not constitute a physical barrier to crime. However, they can act as a catalyst to stimulate crime reduction through a change in perceptions, attitudes, and behavior of residents and potential offenders.

Based on these theoretical perspectives, the ways that CCTV and street lighting can prevent crime are similar. Of course, there are a number of differences between these two situational crime prevention measures. One of these differences rests on the widely held perception that CCTV is far more threatening to law-abiding citizens' civil liberties (Clarke, 2000:104).

It is important to acknowledge that CCTV and improved street lighting could also cause crime to increase. In the case of CCTV, it could give potential victims a false sense of security and make them more vulnerable because they relax their vigilance or stop taking precautions, such as walking in groups at night and not wearing expensive jewelry. It may encourage increased reporting of crimes to the police and increased recording of crimes by the police. If an offender is detected through CCTV, the offense may be recorded and detected at the same time, leading to an increase in police clearance rates. Hence, in order to disentangle

criminal behavior from reporting and recording, both surveys and recorded crime measures are needed in any evaluation. CCTV may also cause crime to be displaced to other locations, times, or victims.

In the case of street lighting improvements, they could, in certain circumstances, increase opportunities for crime. They may bring a greater number of potential victims and potential offenders into the same physical space. Increased visibility of potential victims may allow potential offenders to make better judgments of their vulnerability and attractiveness (e.g., in terms of valuables). Increased social activity outside the home may increase the number of unoccupied homes available for burglary. Increased illumination may make it easier for offenders to commit crimes and to escape. Like CCTV, improved street lighting may also cause crime to be displaced to other areas, times, or victims.

METHODS

This paper brings together the findings of two separate systematic reviews that were carried out to assess the effects of CCTV and improved street lighting on crime (Farrington and Welsh, 2002a, b; Welsh and Farrington, 2002, 2003). It is important to comment briefly on the methodology of systematic reviews in general and the methods employed specifically in the two systematic reviews.

SYSTEMATIC REVIEWS

Systematic reviews use rigorous methods for locating, appraising, and synthesizing evidence from prior evaluation studies, and they are reported with the same level of detail that characterizes high quality reports of original research. According to Johnson et al. (2000:35), systematic reviews "essentially take an epidemiological look at the methodology and results sections of a specific population of studies to reach a research-based consensus on a given study topic." They have explicit objectives, explicit criteria for including or excluding studies, extensive searches for eligible evaluation studies from all over the world, careful extraction and coding of key features of studies, and a structured and detailed report of the methods and conclusions of the review. All of this contributes greatly to the ease of their interpretation and replication by other researchers. It is beyond the scope of this article to discuss all of the features of systematic reviews, but interested readers should consult key reports on the topic (see e.g., Farrington and Petrosino, 2000; Farrington and Welsh, 2001; Johnson et al., 2000).

CRITERIA FOR INCLUSION OF EVALUATION STUDIES

In selecting evaluations for inclusion in the reviews, the following criteria were used:

1. CCTV or improved street lighting was the focus of the intervention. For evaluations involving one or more other interventions, only those evaluations in which CCTV or improved street lighting was the main intervention were included. The determination of the main intervention was based on the author identifying it as such or, if the author did not do this, the importance of CCTV or improved street lighting relative to the other interventions. For a small number of included evaluations with multiple interventions, the main intervention was not identified, but it seemed clear from the report that CCTV or improved street lighting was the most important. It is desirable to include only evaluations where CCTV or improved street lighting was the main intervention, because in other cases, it is impossible to disentangle the effects of these main interventions from the effects of others.

2. There was an outcome measure of crime. The most relevant crime outcomes were violent and property crimes.

3. The evaluation design was of high methodological quality, with the minimum design involving before-and-after measures of crime in experimental and comparable control areas. According to Cook and Campbell (1979) and Shadish et al. (2002), this is the minimum design that is interpretable. Control areas are needed to counter threats to internal validity.

4. The total number of crimes in each area before the intervention was at least 20. The main measure of effect size was based on changes in crime rates between the before and after time periods. It was considered that a measure of change based on an N below 20 was potentially misleading. Also, any study with less than 20 crimes before would have insufficient statistical power to detect changes in crime. The criterion of 20 is probably too low, but we were reluctant to exclude studies unless their numbers were clearly inadequate.

SEARCH STRATEGIES

The following four search strategies were carried out to identify CCTV or improved street lighting evaluations meeting the criteria for inclusion in the two systematic reviews:

1. Searches of online data bases. The following data bases were searched: Criminal Justice Abstracts, National Criminal Justice Reference Service (NCJRS) Abstracts, Sociological Abstracts, Social Science Abstracts (SocialSciAbs), Educational Resources Information Clearinghouse (ERIC), Government Publications Office Monthly Catalog (GPO

Monthly), Psychology Information (PsychInfo), and Public Affairs Information Service (PAIS) International. These data bases were selected because they had the most comprehensive coverage of criminological, criminal justice, and social science literatures. They are also among the top data bases recommended by the Campbell Crime and Justice Coordinating Group, and other systematic reviews of interventions in the field of crime and justice have used them (e.g., Braga, 2001).

For the CCTV systematic review, the following terms were used to search these data bases: closed-circuit television, CCTV, cameras, social control, surveillance, and formal surveillance. When applicable, "crime" was then added to each of these terms (e.g., CCTV and crime) to narrow the search parameters. For the street lighting systematic review, the following terms were used to search the data bases: street lighting, lighting, illumination, and natural surveillance. When applicable, "crime" was then added to each of these terms, again to narrow the search parameters.

2. Searches of literature reviews on the effectiveness of the interventions in preventing crime. For CCTV, the following reviews were searched: Eck (2002), Nieto (1997), Phillips (1999), and Poyner (1993). For street lighting, the following reviews were searched: Eck (2002), Fleming and Burrows (1986), Painter (1996), Pease (1999), Poyner (1993), Ramsay and Newton (1991), and Tien et al. (1979).[1]

3. Searches of bibliographies of CCTV and street lighting reports.

4. Contacts with leading researchers.

Both published and unpublished reports were included in the searches. Furthermore, the searches were international in scope and were not limited to the English language (one non-English language evaluation report is included in the CCTV review).

These search strategies resulted in the identification of 83 evaluations (49 for CCTV and 34 for street lighting). Of these, 76 were obtained and analyzed; the other seven, which may or may not have met the criteria for inclusion, could not be obtained. Of these 76 evaluations, 32 met the criteria for inclusion (19 for CCTV and 13 for street lighting) and 41 did not (25 for CCTV and 16 for street lighting) and thus were excluded from the

1. The review by Tien et al. (1979) identified 103 street lighting projects carried out in the United States in the 1970s, but it only considered that 15 (listed on their pp. 51–54) met their minimum methodological standards. We attempted to obtain 11 of these 15 evaluation reports. For the other four studies (conducted in Baltimore; Chicago; Richmond, Virginia; and Washington, DC), Tien et al. (1979) could not determine from the reports that there was any kind of experimental-control comparison. Hence, we did not attempt to obtain and screen every possible study on street lighting and crime conducted prior to Tien et al. (1979), only studies that conceivably might meet our criteria for inclusion. We did attempt to obtain and screen every possible study conducted after Tien et al.'s (1979) review.

systematic reviews. The three remaining studies (all CCTV) met the criteria for inclusion but did not provide the needed data to be included in the meta-analysis (see below).[2]

FINDINGS

One of the main aims of this paper is to estimate the two interventions' comparative effectiveness in preventing crime,[3] based on the highest quality available research evidence. (Table 1 presents summary information on each of the 32 studies.) To address this and other questions of interest to the present study, results obtained in the included evaluations—on the effects of CCTV and improved street lighting on crime—are analyzed using the statistical technique of meta-analysis.

A meta-analysis is essentially a statistical summary of comparable effect sizes reported in each evaluation. In order to carry out a meta-analysis, a comparable measure of effect size and an estimate of its variance is needed in each program evaluation (Lipsey and Wilson, 2001; Wilson, 2001). In the case of CCTV and street lighting evaluations, the measure of effect size had to be based on the number of crimes in the experimental and control areas before and after the intervention, because this was the only information that was regularly provided in these evaluations. Here, the odds ratio (OR) is used as the measure of effect size. For example, in the Doncaster CCTV evaluation (Skinns, 1998), the odds of a crime after given a crime before in the control area were 2,002/1,780 or 1.12. The odds of a crime after given a crime before in the experimental area were 4,591/5,832 or 0.79. The OR, therefore, was 1.12/0.79 or 1.42. This was statistically highly significant ($p < .0001$).

The OR has a very simple and meaningful interpretation. It indicates the proportional change in crime in the control area compared with the experimental area. An OR greater than 1.0 indicates a desirable effect of the intervention, and an OR less than 1.0 indicates an undesirable effect. In this example, the OR of 1.42 indicates that crime increased by 42% in the control area compared with the experimental area. An OR of 1.42 could also indicate that crime decreased by more than 30% in the experimental area compared with the control area, because the change in the experimental area compared with the control area is the inverse of the OR, or 1/1.42 here.

2. Information on the unobtainable and excluded evaluations is available from the first author.

3. Few of the included studies assessed the impact of the intervention on other outcomes, such as fear of crime.

TABLE 1. SUMMARY OF CCTV AND IMPROVED STREET LIGHTING EVALUATIONS

Author and Publication Date	Place	Period (months) Pre	Post	Other Inter-ventions	Outcome Measure
CCTV in City Centers					
Brown (1995)	Newcastle U.K.	26	15	No	Survey
Brown (1995)	Birmingham U.K.	12	12	No	Records
Short and Ditton (1996)	Airdire U.K.	24	24	No	Records
Skinns (1998)	Doncaster U.K.	24	24	Yes	Records
Armitage et al. (1999)	Burnley U.K.	12	12	No	Records
Farrington et al. (2004)	Cambridge U.K.	11	11	No	Survey/Records
Mazerolle et al. (2002)	Cincinnati N	23	6	No	Records
Mazerolle et al. (2002)	Cincinnati H	23	4	No	Records
Mazerolle et al. (2002)	Cincinnati F	24.5	3.5	No	Records
Street Lighting in City Centers					
Poyner and Webb (1997)	Birmingham U.K.	12	12	No	Records
Atlanta RC (1974)	Atlanta	12	12	No	Records
DIFL (1974)	Milwaukee	12	12	No	Records
Wright et al. (1974)	Kansas City MO	12	12	No	Records
Sternhell (1977)	New Orleans	51	29	No	Records
CCTV in Residential/Public Housing					
Musheno et al. (1978)	New York	3	3	No	Survey

Street Lighting in Residential/Public Housing					
Shaftoe (1994)	Bristol U.K.	12	12	No	Records
Painter and Farrington (1997)	Dudley U.K.	12	12	No	Survey/SR
Painter and Farrington (1999)	Stoke U.K.	12	12	No	Survey
Inskeep and Goff (1974)	Portland OR	6 or 11	6 or 11	No	Records
Harrisburg PD (1976)	Harrisburg	12	12	No	Records
Lewis and Sullivan (1979)	Fort Worth	12	12	No	Records
Quinet and Nunn (1998)	Indianapolis	6-9	6-9	Yes	Records
CCTV in Car Parks					
Poyner (1991)	Guildford U.K.	24	9	Yes	Records
Tilley (1993)	Hartlepool U.K.	15	30	Yes	Records
Tilley (1993)	Bradford U.K.	12	12	Yes	Records
Tilley (1993)	Coventry U.K.	8/16	8/16	Yes	Records
Sarno (1996)	Sutton U.K.	12	12	Yes	Records
Street Lighting in Car Parks					
Poyner (1991)	Dover U.K.	24	24	Yes	Records
CCTV in Public Transportation					
Burrows (1979)	Underground S	12	12	Yes	Records
Webb and Laycock (1992)	Underground N	46	26	Yes	Records
Webb and Laycock (1992)	Underground C	28	32	Yes	Records
Grandmaison and Tremblay (1997)	Montreal	18	18	No	Records

NOTES: Cincinnati: N = Northside, H = Hopkins Park, F = Findlay Market; RC = Regional Commission; DIFL = Department of Intergovernmental Fiscal Liaison; SR = self-reports; PD = police department; 8/16 = 8 for experimental, 16 for control (in both pre and post); Underground: S = Southern sector, N = Northern line, C = Central.

The OR is calculated from the following table:

	Before	After
Experimental	a	b
Control	c	d

a, b, c, and d are the numbers of crimes

OR = ad/bc.

The variance of OR is calculated from the variance of LOR (the natural logarithm of OR):

V(LOR) = 1/a + 1/b + 1/c + 1/d

In order to produce a summary effect size in a meta-analysis, each effect size is weighted according to the inverse of the variance. This was another reason for choosing the OR, which has a known variance (Fleiss, 1981:61–67).[4]

We also investigated the important issues of displacement of crime and diffusion of crime prevention benefits. Displacement is often defined as the unintended increase in targeted crimes in other locations following from the introduction of a crime reduction scheme.[5] Five different forms of displacement have been identified by Reppetto (1976): temporal (change in time), tactical (change in method), target (change in victim), territorial (change in place), and functional (change in type of crime). Diffusion of benefits is defined as the unintended decrease in crimes following from a crime reduction scheme, or the "complete reverse" of displacement (Clarke and Weisburd, 1994).

IS CCTV OR IMPROVED STREET LIGHTING MORE EFFECTIVE?

Separate meta-analyses of the 19 CCTV evaluations and the 13 improved street lighting evaluations provided evidence that both interventions were equally effective in reducing (total) crime. In the case of CCTV, the weighted mean OR was 1.27 (95% confidence interval

4. This estimate of the variance is based on the assumption that total numbers of crimes (a, b, c, d) have a Poisson distribution. Thirty years of mathematical models of criminal careers (see, e.g., Blumstein et al., 1986; Piquero et al., 2003) have been dominated by the assumption that the commission of crimes can be accurately modeled by a Poisson process. For a more detailed discussion of the variance in this case, see Farrington and Welsh (2004). In the 32 evaluations, we empirically estimated the ratio of the variance to the mean, based on the before and after number of crimes in control areas. The average was 1.21, which is close to the Poisson figure of 1. (The geometric mean was used as an average because of the ratio variables.)

5. For a discussion of "benign" or desirable effects of displacement, see Barr and Pease (1990).

1.11–1.46, p = .0004). This means that crimes increased by 27% after CCTV in control areas compared with experimental areas, or, conversely, crimes decreased by 21% in experimental areas compared with control areas. In the case of improved street lighting, the weighted mean OR was 1.28 (95% confidence interval 1.11–1.48), which was also a highly significant effect (p = .0008). This means again that crimes increased by 28% after improved street lighting in control areas compared with experimental areas, or, conversely, crimes decreased by 22% in experimental areas compared with control areas.[6]

SETTING

All of the 19 CCTV evaluations and the 13 improved street lighting evaluations were carried out in one of four settings: city center, residential or public housing, car parks, or public transportation (see Table 2). With the exception of CCTV in residential/public housing (only one study), all of the settings in which either CCTV or improved street lighting schemes could be evaluated showed desirable effects on crime, as measured by the weighted mean OR. CCTV in car parks had the largest effect on crime, with an overall OR of 1.77 (95% confidence interval 1.37–2.28, p < .0001). In city centers, improved street lighting (OR = 1.47, p = .022) was more effective than CCTV (OR = 1.15, ns).

CRIME TYPE

Table 3 shows that CCTV and street lighting were more effective in reducing property crimes (OR for CCTV = 1.54, p < .0001; OR for lighting = 1.27, p = .019) than violent crimes (OR for CCTV = 1.05, ns; OR for lighting = 1.15, ns). Property crimes included burglary, vehicle crimes, and theft, and violent crimes included robbery and assault.

COUNTRY COMPARISON

Of the 19 CCTV evaluations, 14 were from Britain and the other 5 were from North America (4 from the United States and 1 from Canada). Of the 13 improved street lighting evaluations, 8 were from the United States and the other 5 were from Britain. As illustrated in Table 4, when the pooled meta-analysis results for each intervention were disaggregated by country, CCTV and improved lighting were more effective in reducing crime in Britain (OR for CCTV = 1.43, p = .0003; OR for lighting = 1.63, p = .0002) than in North America (OR for CCTV = 0.99, ns; OR for lighting = 1.12, ns).

6. These effect size estimates are based on a random effects model, which is appropriate when effect sizes are significantly heterogeneous (as here). Our previous meta-analyses were based on a fixed effects model.

TABLE 2. META-ANALYSIS OF CCTV AND IMPROVED STREET LIGHTING EVALUATIONS, BY CONTEXT OF INTERVENTION

Context of Intervention	CCTV (N = 19)	Street Lighting (N = 13)
City Center	OR = 1.15 (0.96 - 1.37) (n = 9; Newcastle, Birmingham, Airdrie, Doncaster, Burnley, Cambridge, Cincinnati N, Cincinnati H, Cincinnati F)	OR = 1.47* (1.06 - 2.04) (n = 5; Birmingham, Atlanta, Milwaukee, Kansas City, New Orleans)
Residential/ Public Housing	OR = 0.89 (0.42 - 1.91) (n = 1; New York)	OR = 1.22* (1.01 - 1.47) (n = 7; Bristol, Dudley, Stoke, Portland, Harrisburg, Fort Worth, Indianapolis)
Car Parks	OR = 1.77* (1.37 - 2.28) (n = 5; Guildford, Hartlepool, Bradford, Coventry, Sutton)	OR = 1.14 (0.62 - 2.08) (n = 1; Dover)
Public Transportation	OR = 1.30 (0.87 - 1.96) (n = 4; Underground S, Underground N, Underground C, Montreal)	n.a.

* $p < .05$.
NOTES: OR = weighted mean odds ratio (and 95% confidence interval); Cincinnati: N = Northside, H = Hopkins Park, F = Findlay Market; Underground: S = Southern sector, N = Northern line, C = Central; n.a. = not available.

DISPLACEMENT OF CRIME AND DIFFUSION OF CRIME PREVENTION BENEFITS

As shown in Table 5, 11 of the 19 CCTV studies and 12 of the 13 street lighting studies measured displacement of crime, diffusion of crime prevention benefits, or both. Across the CCTV studies, mixed results were found for territorial displacement and diffusion of benefits. For example, in the case of the seven city center CCTV studies that measured displacement, diffusion, or both (Newcastle to Cincinnati F; see Table 5), four studies reported at least some evidence of territorial displacement and

TABLE 3. META-ANALYSIS OF CCTV AND IMPROVED STREET LIGHTING EVALUATIONS, BY CRIME TYPE

Crime Type	CCTV	Street Lighting
Violent (total)	OR = 1.05 (0.83 - 1.34) (n = 6; Airdrie, Burnley, Cambridge, Underground N, Underground C, Montreal)	OR = 1.15 (0.90 - 1.48) (n = 9; Bristol, Dudley, Stoke, Atlanta, Milwaukee, Portland, Kansas City, Harrisburg, New Orleans)
Property (total)	OR = 1.54* (1.24 - 1.91) (n = 11; Newcastle, Airdrie, Burnley, Underground S, Underground C, Montreal, Guildford, Hartlepool, Bradford, Coventry, Sutton)	OR = 1.27* (1.04 - 1.55) (n = 11; Dover, Bristol, Birmingham, Dudley, Stoke, Atlanta, Milwaukee, Portland, Kansas City, Harrisburg, New Orleans)

*$p < .05$.
NOTES: OR = weighted mean odds ratio (and 95% confidence interval); Underground: S = Southern sector, N = Northern line, C = Central.

three studies reported at least some evidence of diffusion of benefits. Different results were found for the street lighting studies. Only three of these studies reported some or possible evidence of territorial displacement and the other nine reported no evidence of displacement, with two

TABLE 4. META-ANALYSIS OF CCTV AND IMPROVED STREET LIGHTING EVALUATIONS, BY COUNTRY

Country	CCTV	Street Lighting
United Kingdom	OR = 1.43* (1.18 - 1.74) (n = 14)	OR = 1.63* (1.26 - 2.12) (n = 5)
U.S. and Canada	OR = 0.99 (0.95 - 1.02) (n = 5)	OR = 1.12 (0.97 - 1.27) (n = 8)
Total	OR = 1.27* (1.11 - 1.46) (N = 19)	OR = 1.28* (1.11 - 1.48) (N = 13)

*$p < .05$.
Note: OR = weighted mean odds ratio (and 95% confidence interval).

of these studies (Birmingham and Stoke) also reporting at least some evidence of diffusion.

TABLE 5. FINDINGS ON DISPLACEMENT OF CRIME AND DIFFUSION OF CRIME PREVENTION BENEFITS FOR CCTV AND IMPROVED STREET LIGHTING EVALUATIONS

	CCTV (N = 19)
Place	Findings
Newcastle U.K.	Some displacement and diffusion occurred
Birmingham U.K.	Displacement occurred
Airdrie U.K.	Diffusion occurred
Doncaster U.K.	No displacement occurred
Burnley U.K.	Diffusion occurred
Cambridge U.K.	Not measured
Cincinnati N	Little or no displacement occurred
Cincinnati H	Not measured
Cincinnati F	Some displacement occurred
New York	Not measured
Guildford U.K.	Diffusion occurred
Hartlepool U.K.	Displacement occurred
Bradford U.K.	Not measured
Coventry U.K.	Not measured
Sutton U.K.	Not measured
Underground S	Some displacement occurred
Underground N	Diffusion occurred
Underground C	Not measured
Montreal	Not measured
	Street Lighting (N = 13)
Place	Findings
Birmingham U.K.	No displacement occurred, some diffusion occurred
Atlanta	No displacement occurred
Milwaukee	Some displacement occurred
Kansas City MO	Some displacement occurred
New Orleans	No displacement occurred
Bristol U.K.	Not measured
Dudley U.K.	No displacement occurred
Stoke U.K.	No displacement occurred, diffusion occurred
Portland OR	No displacement or diffusion occurred
Harrisburg	No displacement occurred
Fort Worth	Possible displacement occurred
Indianapolis	No displacement occurred
Dover U.K.	No displacement occurred

NOTES: CCTV and street lighting studies, respectively, are listed in the same order as they appear in Table 1. Cincinnati: N = Northside, H = Hopkins Park, F = Findlay Market; Underground: S = Southern sector, N = Northern line, C = Central.

In order to investigate displacement of crime and diffusion of crime prevention benefits, the minimum design should involve one experimental area, one adjacent area, and one nonadjacent comparable control area. If crime decreased in the experimental area, increased in the adjacent area, and stayed constant in the control area, this might be evidence of displacement. If crime decreased in the experimental and adjacent areas and stayed constant or increased in the control area, this might be evidence of diffusion of benefits. Unfortunately, few CCTV or street lighting studies used this minimum design. Instead, most had an adjacent control area and the remainder of the city as another (noncomparable) control area. Because of this, any conclusions about displacement or diffusion effects of CCTV and street lighting seem premature at this point in time.

DISCUSSION AND CONCLUSION

The findings and a number of key issues facing these two forms of surveillance merit further discussion.

POLICY IMPLICATIONS

On the basis of the highest quality available research evidence, both CCTV surveillance cameras and improved street lighting represent effective situational measures for reducing crime. For city managers, business owners, or others, this may be useful information if a decision needs to be made about implementing one or the other measure. (Hopefully, as Moore [2002] argued for in another context, a cost-effectiveness analysis or comparative cost-benefit analysis would also be carried out to inform this decision.) But for some, this information may be too limited.

What may be more helpful to practitioners and more useful for policy discussions is information about the specific conditions under which improved street lighting and CCTV are most effective in reducing crime. The present research showed that CCTV and improved street lighting were effective in reducing (total) crime, with one exception, in each of the four settings they were evaluated, with the largest effect on crime being for CCTV in car parks. And in the one setting (city centers) that CCTV could be compared with improved street lighting, the latter was found to be more effective in reducing (total) crime. Although limited to the one setting of city centers, arguably one of the most important public spaces to implement crime prevention programs, this finding provides further direction for the allocation of scarce public resources.

Also of importance was evidence that showed that CCTV and improved lighting were more effective in reducing property crimes than in reducing violent crimes. Regular crime analysis by the police, such as used in Compstat, could be used to identify those places that are at greatest risk

for property crimes, particularly vehicle crimes, which, in turn, could be used to guide the deployment of CCTV surveillance cameras or improved lighting. The advent of mobile and removable CCTV units may make this a more feasible and perhaps less costly option.

Another important issue from the findings of these two situational crime prevention measures is the potential benefit that may come from combining them. As discussed above, both measures may reduce crime through a number of similar mechanisms, for example, by increasing offenders' perceived risks associated with committing a crime or strengthening community cohesion and social control. They may be more effective with property crimes because such crimes are more "rational" than violent crimes. Arguably, there is no conflict in implementing these two measures alongside each other.[7] So, is there empirical support for combining these two interventions?

Five of the 19 CCTV evaluations (Underground N, Guildford, Bradford, Coventry, and Sutton; see Table 1) used improved lighting as a secondary intervention. (None of the 13 improved street lighting evaluations used CCTV as a secondary intervention.) A meta-analysis of these five evaluations provided evidence that the combination of CCTV and improved lighting can be highly effective in reducing crime, with a weighted mean OR of 1.65 (95% confidence interval 1.24–2.21, $p = .0007$). A meta-analysis of the other 14 CCTV evaluations (four of these used other interventions, such as security officers and notices of CCTV) produced a weighted mean OR of 1.20 (95% confidence interval 1.04–1.39, $p = .016$). From these results, it can be seen that the combination of CCTV and improved lighting (as primary and secondary measures, respectively) was the more effective form of surveillance in reducing crime, and the difference was borderline significant ($p = .054$).[8] It is important to note, however, that four of these five evaluations took place in car parks and

7. Some may view the combination of CCTV surveillance cameras and street lighting as trivial, based on the notion that CCTV requires natural or artificial lighting to work. This is not the case. Technological innovations in CCTV cameras, such as infrared, allow cameras to work without other lighting. Thus, improved street lighting and CCTV should be viewed as distinct interventions. There may be a conflict between the two measures if improved lighting increases community cohesion and CCTV cameras give the impression that a place is unsafe, thus detracting from social cohesion because residents are afraid to go there.

8. The significance of a difference between two ORs was tested by comparing LORs (natural logarithms of ORs). The formula for the variance of LOR, V(LOR), is given above. In this example, LOR for the five CCTV plus lighting evaluations = .50, SE = .15; LOR for the other 14 CCTV evaluations = .18, SE = .075. The pooled SE was .17, giving $z = .32/.17 = 1.93$, $p = .054$. No other comparison of two ORs was statistically significant.

only measured the schemes' effect on vehicle crimes. Therefore, the powerful result evidenced from combining CCTV surveillance cameras and improved street lighting may be limited to targeting vehicle crimes in car parks.

LESSONS FOR AMERICA?

Another interesting finding to emerge from the present research is that both forms of surveillance were found to be far more effective in reducing crime in Britain than in America.[9] What might account for this? Or, more importantly, what lessons can be drawn from the British studies to help improve the effectiveness of CCTV and improved street lighting in America? There were some differences in key characteristics between the British and American CCTV schemes and between the British and American improved street lighting schemes, which may help to address these questions.

First, the average follow-up period of the five American CCTV schemes was substantially shorter than for the 14 British CCTV schemes: 6.9 months versus 17.7 months. (Four of the American evaluations had the shortest follow-up periods of all 19 CCTV evaluations, ranging from a low of three months to a high of six months.) Because of the short follow-up periods in the American studies, it is possible that the CCTV schemes were not given enough time to produce a clear effect on crime, either desirable or undesirable (all five of the American studies showed evidence of either a null or uncertain effect on crime). Longer follow-up periods, as in the majority of the British studies, seem to be warranted for future American CCTV evaluations. No difference was found in the average follow-up period for the British and American improved street lighting evaluations (14.4 months vs. 13.6 months, respectively).

Second, and perhaps most importantly, not one of the five American CCTV evaluations used other interventions alongside CCTV, although nine of the 14 British schemes used one or more other type of intervention, such as improved lighting or police patrols. If the five car park schemes are removed, because all of them were carried out in Britain and involved other interventions, this leaves four (three with a desirable and one with an undesirable effect on crime) out of nine British studies that used other interventions. It is possible that the absence of policing or other situational crime prevention measures in the American CCTV schemes may be a contributing factor to their overall poor effect in reducing crime; for example, CCTV on its own may not represent a sufficient

9. Reference is hereafter made to America because, with the exception of the Grandmaison and Tremblay (1997) CCTV evaluation, all of the CCTV and improved street lighting evaluations were conducted in either Britain or America.

deterrent threat to influence an offender's decision-making process to commit a crime or not. No difference was found between the British and American improved street lighting schemes on their use of other interventions.

Another important issue that may be a contributing factor to the difference in effectiveness between the British and American schemes, both for CCTV and improved street lighting, is cultural context. In Britain, there is a high level of public support for the use of CCTV cameras in public settings to prevent crime (Norris and Armstrong, 1999:60–62; Phillips, 1999:139–140), whereas in America the public is less accepting of and more apprehensive of Big Brother implications arising from this surveillance technology (Murphy, 2002). In America, resistance to the use of CCTV in public space also takes the form of legal action and constitutional challenges under the U.S. Constitution's Fourth Amendment prohibition against unreasonable searches and seizures (Nieto, 1997:1).

It could very well be that the poor showing of the American CCTV schemes was due in part to a lack of public support (and maybe even political support) for the schemes, which, in turn, may have resulted in cuts in program funding, the police assigning lower priority to the schemes, or attempts to discourage desirable media coverage, for example. Each of these could potentially undermine the effectiveness of CCTV schemes. In contrast, the British Home Office, who funded many of the British evaluations, would have liked to find that CCTV was effective.

Although cultural context could play a role in the differential effectiveness of street lighting in the two countries (to our knowledge there have been no recent surveys of the public in either country), it is our opinion that this is not likely. This is mainly because this form of surveillance is viewed as having few if any perceived harmful social consequences, unlike CCTV.

Lastly, for improved street lighting, there may be something of an age effect that has contributed to the difference in effectiveness between the British and American schemes. With the exception of one of the eight American street lighting evaluations, all of them were carried out at least 10 to 15 years earlier than the first British street lighting evaluation. Could it be that the British street lighting evaluations drew upon the knowledge gleaned from the individual American evaluations and the detailed review by Tien et al. (1979), and that this played some role in the effectiveness of the British lighting schemes? This is possible, because there was a great awareness of this American research, which is evidenced in British-based reviews of the literature (Painter, 1996; Ramsay and Newton, 1991) and in some of the British lighting studies included here.

Another factor that may have contributed to the difference in effectiveness between the American and British street lighting schemes is the possibility that offenders during the 1990s (in the case of the British studies) may have been influenced by different factors compared with those over a decade ago.

OTHER POTENTIAL BENEFITS

One of the issues that this paper was not able to explore was that CCTV and improved street lighting may produce other important benefits beyond a reduction in crime rates. Improvements in street lighting may also provide benefits (during nighttime hours) in the form of, for example, increased pedestrian and traffic safety and increased usage of public parks and other recreational areas.

Compared with street lighting, there are, arguably, many more potential benefits associated with CCTV. For the police, the potential benefit of CCTV in reducing crime by deterring offenders from committing an illegal activity may be much lower on its list of priorities than the apprehension of suspects that were caught on camera committing a crime; police officer safety (by installing CCTV cameras in patrol cars to record the events of road-side stops, for example); or the prevention of terrorist activities, whether it be through the identification of suspicious persons or packages that prevent the act from taking place or to aid in the identification, apprehension, and conviction of suspects after the act. It was mainly this latter function of CCTV that was used to justify, at least in the beginning, its widespread implementation in public and private space throughout Britain (Rosen, 2001). In America, the terrorist activities perpetrated on September 11, 2001, have contributed to an increased use of CCTV surveillance cameras in this country (Clymer, 2002; Murphy, 2002).

Improved street lighting is an effective form of surveillance to reduce crime in public space and it has few if any perceived harmful social consequences (unlike CCTV), and it may attract less public resistance than CCTV surveillance cameras (in America at least). However, the research evidence shows that CCTV is also an effective form of surveillance for preventing crime and that it has many other potential benefits. As there is also evidence of the effectiveness of combining these two forms of surveillance, perhaps it is time for researchers and policy-makers to look more closely at how to maximize their shared effectiveness in reducing crime and to capitalize on their different qualities.

DIRECTIONS FOR RESEARCH

Future research should begin with attention to the methodological rigor of the evaluation designs. The use of a comparable control group by all of

the 32 included evaluations went some way toward ruling out some of the major threats to internal validity, such as selection, maturation, history, and instrumentation. The effect of CCTV and street lighting on crime can also be investigated after controlling (e.g., in a regression equation) not only for prior crime but also for other community-level factors that influence crime, such as neighborhood poverty and poor housing. Another possible research design is to match two areas and then to choose one at random to be the experimental area. Ideally, several experimental areas and several comparable adjacent and control areas would be included. Adjacent and nonadjacent areas are needed to test hypotheses about displacement and diffusion of benefits.

Future research should also investigate some of the more subtle qualitative aspects of street lighting, such as lighting levels and styles. In one study on street lighting in Toronto, it was found that high-pressure sodium lighting was identified with higher fear levels among women respondents (METRAC, 1989). This finding led the City of Toronto to switch thousands of their public street lights to metal halide.

Research is also needed on the monetary costs and benefits of CCTV and street lighting programs. As noted above, it would be useful to measure which of these two interventions is more cost-effective in preventing crime. It would also be useful to measure if a program that combined both CCTV and street lighting was more cost-beneficial than either of the two implemented on their own. We had hoped to be able to examine these issues, but it was not possible because only 3 of the 32 included studies (Painter and Farrington, 1997, 1999; Skinns, 1998) carried out a cost-benefit analysis. In each case, benefits from reduced crime substantially outweighed program costs. Future cost-benefit analyses of CCTV and street lighting programs should take account of any displacement of crime or diffusion of crime prevention benefits. Although there remain a number of other issues pertaining to cost-benefit analysis of situational crime prevention in need of examination (see Roman and Farrell, 2002), our previous work (Welsh and Farrington, 1999, 2000) has shown that situational crime prevention generally is an economically efficient strategy.

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